Lecture 24

Tuesday, 11 April 2023 1:35 PM

Demo: RDE ; Au disk (WE)

Hg/Hg_2SO_4 RE (0.62 V vs. SHE)

Pt wire CE

0.1 M K_2SO_4 (+ H_2SO_4) → pH ≈ 4

Relevant Rxns:

(1) Hydrogen evolution reaction (HER)

$2H^+ + 2e^- \rightarrow H_2$; $U^\circ = 0$ vs. SHE @ pH = 0

$U = 0$ V vs. RHE ← built-in pH correction (shifts w/ pH of solution)

@ pH ≈ 4, electrolyte is proton-limited

HER still occurs when we run out of protons

via $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ $U^\circ = -0.828$ V vs. SHE @ pH = 14

$U = 0$ V vs. RHE

(2) Oxygen Reduction Reaction (since we are not purging w/ Ar so O_2 will also react @ electrode)

$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ $U^\circ = 1.23$ V vs. SHE @ pH = 0

$U = 1.23$ V vs. RHE

$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ $U^\circ = 0.401$ V vs. SHE @ pH = 14

$U = 1.23$ V vs. RHE

Nernst

$V_{applied} = V_{RHE} - V_{RE} - (0.0591\ V \times pH)$

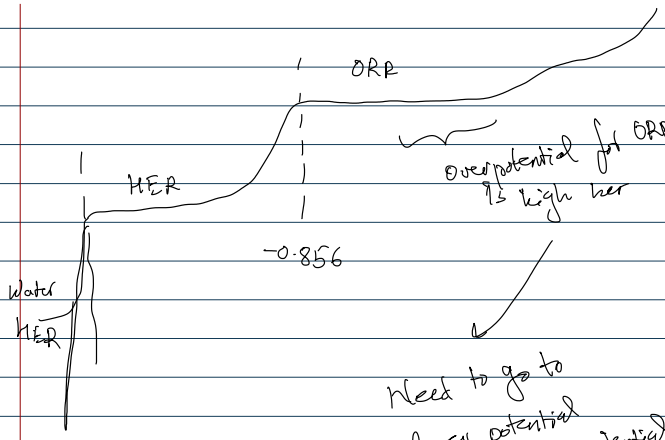
$V_{applied} = 0 - 0.62 - 0.0591\,(4) = -0.856$ V vs.

@ 0 V vs. RHE, $\eta_{HER} = V_{applied} - U = 0$ We are giving it no "extra" potential (i.e. no reaction should be driven)

$\eta_{ORR} = -1.23$ V ← very large driving force for this to happen so @ $V_{applied} = 0$ it will be dominating

We want current density on CE to be very small (thus high SA)

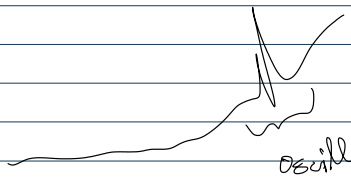
CV → linear change b/w two vertexes



Overpotential for ORR is high here

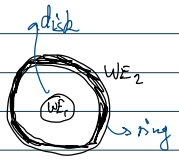
Need to go to lower potential to have overpotential for HER (But ORR is still happening as well)

Diff coeff → tells you spacing b/w them (resulting in diff. slopes)



Oscillation is due to potentiostat switching it's I-range (since it is Auto)

Rotating Ring Disk Electrode



→ "Generator - collector" mode

↳ Redox species generated @ disk, and that species does not exist in bulk electrolyte

↳ Ring potential is held s.t. generated specie conc. is zero @ ring surface

If not known, need to do sweep & measure i to see where / what could be possible

For, $C_{ring} = 0$, the MT equations solved to yield i_{ring} as function of i_{disk}

$$i_{ring} = \frac{nF}{s} D \int_{r_2}^{r_3} \left.\frac{\partial c}{\partial y}\right|_{y=0} r \cdot dr$$

Solution provides:

$$i_{ring} = -N_c \, i_{disk}$$

↳ collection (only a function of

$i_{ring} = -N_c\, i_{disk}$

↳ collection efficiency (fraction of species generated @ disk that reacts @ ring) (only a function of r_1, r_2, r_3)

↳ No function of Ω, D, C_∞

How to maximize N_c?

+ Small r_2 and + Large $(r_3 - r_2)$ } Small gap & large ring

[Since you want to collect all the species formed on the disk which are moving outwards]

→ "Detection Mode"

↳ Instead of generated species reacting @ ring; avg conc of species @ ring induces change in ring equil. potential

↳ This can be useful to detect surface pH changes if ring is potentiometric pH sensor

↳ Hydrodynamics are the same, but now we are not reacting on ring

$$(C_{ring} - C_\infty) = N_D (C_{disk} - C_\infty)$$

↳ detection efficiency (fraction of species generated @ disk, detected as surf conc. @ ring

↳ only depends on r_1, r_2, r_3

Maximizing N_D?

Small ring — Don't want to "average" out the measurement